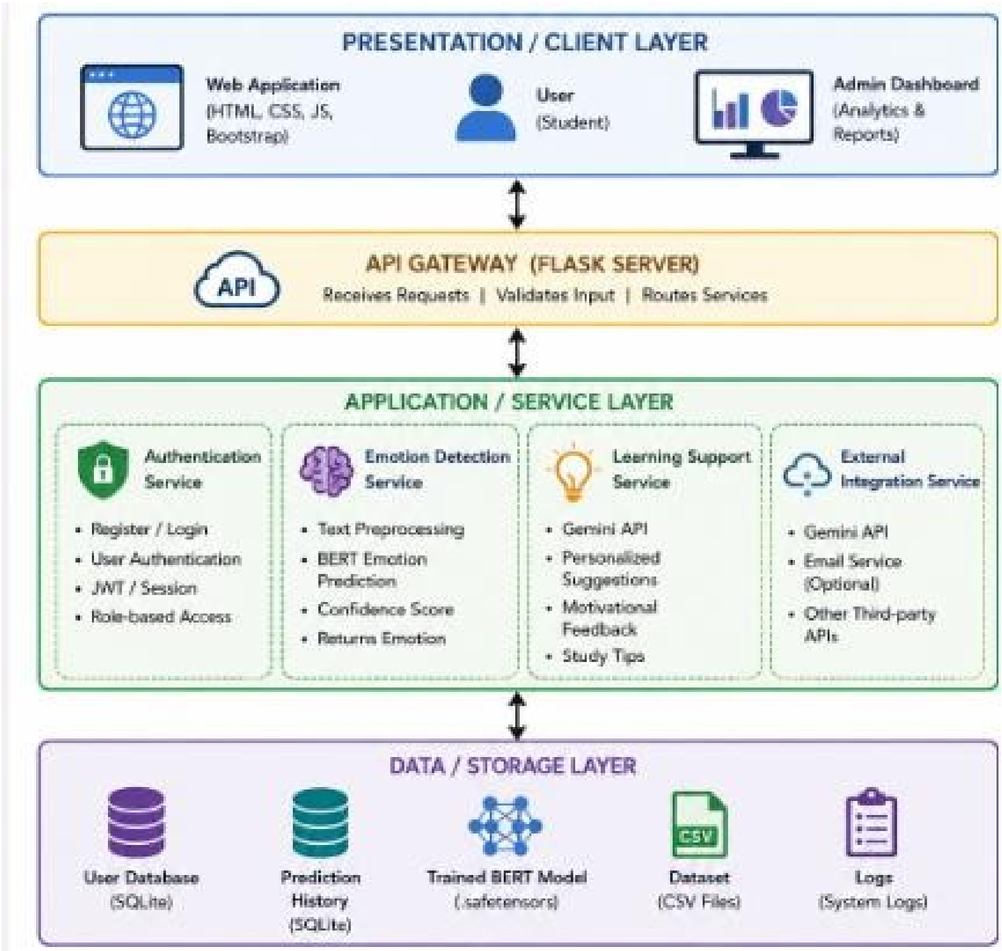
Solution Architecture

Date	15 July 2026
Team ID	XXXX
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Solution Architecture Diagram:

The **Emotion Detection Learning Support Engine** follows a layered architecture consisting of the **Presentation Layer, API Gateway, Application/Service Layer, and Data/Storage Layer**. The system allows users to enter text through a web interface, where the input is securely processed by the backend. The application preprocesses the text, predicts the user's emotion using a fine-tuned BERT machine learning model, and generates personalized learning support using the Gemini API. All prediction results, user information, and logs are stored in the database for future analysis. This architecture ensures scalability, security, maintainability, and efficient real-time emotion detection.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides a user-friendly web interface where users can register, log in, enter text, and view detected emotions along with personalized learning support	HTML, CSS, JavaScript, Bootstrap
API Gateway	Receives requests from the frontend, validates them, and routes them to the appropriate backend services.	Flask (Python)
Authentication Service	Handles user registration, login, logout, JWT/session authentication, and role-based access.	Flask, Python
Emotion Detection Service	Cleans and preprocesses text, loads the trained BERT model, predicts the user's emotion, and returns confidence scores.	Python, Transformers (BERT), PyTorch, scikit-learn
Learning Support Service	Uses the detected emotion to generate personalized motivational messages, study tips, and learning recommendations through the Gemini API.	Gemini API, Python
External Integration Service	Integrates with external AI services and optional email notification services.	Gemini API, REST API
Data / Storage Layer	Stores user details, prediction history, trained model files, datasets, and application logs.	SQLite, CSV Files, Safetensors, File System